Hospital Communication Network Using Prim's Algorithm

1. Problem Statement

To represent hospital blocks as vertices and possible fiber-optic cable connections as edges with installation costs. Apply Prim's Algorithm to determine the Minimum Spanning Tree (MST), which connects all hospital blocks with minimum total installation cost.

2. Objectives

1. To represent hospital blocks using a weighted graph.
2. To identify possible fiber-optic connections between hospital blocks.
3. To apply Prim's Algorithm to find the Minimum Spanning Tree.
4. To minimize the total cost of installing fiber-optic cables.
5. To provide reliable communication between different hospital departments.

3. Algorithm

Prim's Algorithm

1.Start with any one hospital block as the initial vertex.

2. Mark the selected vertex as visited.

3. Find the minimum-cost edge connecting a visited vertex to an unvisited vertex.

4. Select that edge and add the new vertex to the network.

5. Repeat the process until all hospital blocks are connected.

6. Add the costs of all selected edges to calculate the minimum total installation cost.

7. Display the selected connections and the minimum total cost.

4. Advantages

1. Minimizes the total fiber-optic installation cost.

2. Connects all hospital blocks into a single network.

3. Avoids unnecessary cable connections.

4. Improves communication between hospital departments.

5. Provides an efficient solution for network planning.

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5. Applications

1. Connecting hospital departments such as ICU, laboratories, and emergency wards.

2. Establishing communication between hospital buildings.

3. Sharing electronic health records across departments.

4. Connecting medical equipment and hospital information systems.

5. Planning communication networks for large hospitals and healthcare campuses.

6. Conclusion

Prim's Algorithm efficiently determines the Minimum Spanning Tree for a hospital communication network. By selecting the lowest-cost connections needed to connect all hospital blocks, it reduces the total fiber-optic installation cost while providing network connectivity.